NUERS official project branding

NUERS.NET — Comprehensive BIR EIS Integration & Functional Specification

Version 2.0 | October 2026 | For product, engineering, QA, compliance, and security review. Covers every admin navigation page, workflows, forms, modals, data, architecture and BIR EIS integration.

Regulatory boundary: This document is a NUERS project specification, not BIR-issued documentation or proof of accreditation. RMC 110-2026 is the user-provided regulatory source; official EIS API schemas and technical specifications must be separately obtained and verified. All illustrated UI data, BIR badges and example identifiers are mock data.

1. Regulatory framework and why each system capability exists

Requirement	NUERS response / source
ESP accreditation	Accreditation tracking, approved version/environment, controlled production gate. RMC pp.3,7-8.
Taxpayer authority	Verify registration and taxpayer PTI/PTT permissions; ESP approval is not taxpayer permission. RMC pp.3-5,8.
Issuance and delivery	Structured e-invoices, preserved audit trail, near-real-time BIR delivery within applicable 3-calendar-day limit. RMC pp.2,4.
Integrity and reconciliation	Accurate, complete, authentic, sequenced, nonduplicated transactions and reconciled acknowledgments. RMC pp.8-9.
Incident and security	Security assessments, certifications, critical incident reporting within 24 hours. RMC pp.6,9.
Taxpayer data rights	Ownership, continued access, export and migration without vendor lock-in. RMC pp.9-10.
Business continuity	Backup, DR, failover, manual invoicing recovery and linkage. RMC pp.8-9.

The circular does not provide authoritative production EIS endpoint URLs, API credentials, payload schemas, exact response codes, tax computation rules, or a definitive receipt-vs-invoice classification. These are external regulatory dependencies; developers must not invent them.

2. Platform architecture and transaction lifecycle

Flow: POS / ERP / e-commerce / NUERS Portal -> WAF + authenticated tenant API -> taxpayer eligibility + PTI/PTT gate -> canonical invoice validation -> immutable invoice ledger and transactional outbox -> durable queue -> BIR EIS adapter -> acknowledgment storage -> reconciliation -> taxpayer portal / reports / audit.

State machine: Draft -> Validated -> Issued -> Queued -> Transmitting -> Submitted -> Acknowledged / Rejected / Uncertain. Acknowledged means confirmed by an authoritative BIR response, not merely HTTP 200. Uncertain requires inquiry and reconciliation before any replay. Corrections are controlled events, never silent mutation of issued invoices.

Idempotency: each source has stable source ID + tenant + branch + document type; requests with the same idempotency key return the original result. Persist invoice and outbox atomically. Queue consumers must tolerate duplicate delivery.

Deadline management: submit in real/near-real time; track transaction time, attempt time, BIR acknowledgment and three-calendar-day compliance deadlines. Escalate before breach, and preserve evidence of BIR outages and recovery.

3. Core data entities

organizations, branches, users, roles, taxpayer_verifications, pti_ptt_authorizations, esp_accreditations, api_clients, api_keys, documents, document_lines, tax_breakdowns, invoice_sequences, transaction_events, outbox_events, transmission_attempts, bir_acknowledgments, reconciliation_cases, exceptions, audit_events, incident_reports, exports, legal_holds, backups, change_requests, support_tickets.

Every transactional record includes immutable ID, tenant and branch ID, timestamps in UTC with Asia/Manila display, canonical schema version, status history, actor/system principal and source correlation ID. Encrypt personal information and isolate tenants server-side.

4. User roles and permissions

Super Admin: global operations under least privilege; Compliance Officer: accreditation and PTI/PTT review; Transmission Operator: retry and queue management without altering issued data; Taxpayer Admin: own branches and users; Taxpayer

Operator: authorized invoice issuance; Auditor: read-only evidence; Security Officer: incidents and access reviews. Require MFA for privileged roles and two-person approval for sensitive overrides.

5. Screen specification: Dashboard

Design reference only; mock statuses and sample business data are not production evidence.

Purpose and EIS justification

Operational overview, KPIs, gateway status, alerts and navigation.

Controls, forms and modals

Date and tenant filters; metric drilldown; alert detail drawer.

Standard form requirements: explicit required fields, server-side validation, contextual help, disabled submit while saving, meaningful error states, accessible keyboard focus and cancellation without partial writes. Confirmation dialogs show affected tenant/document, operation, risk and reason.

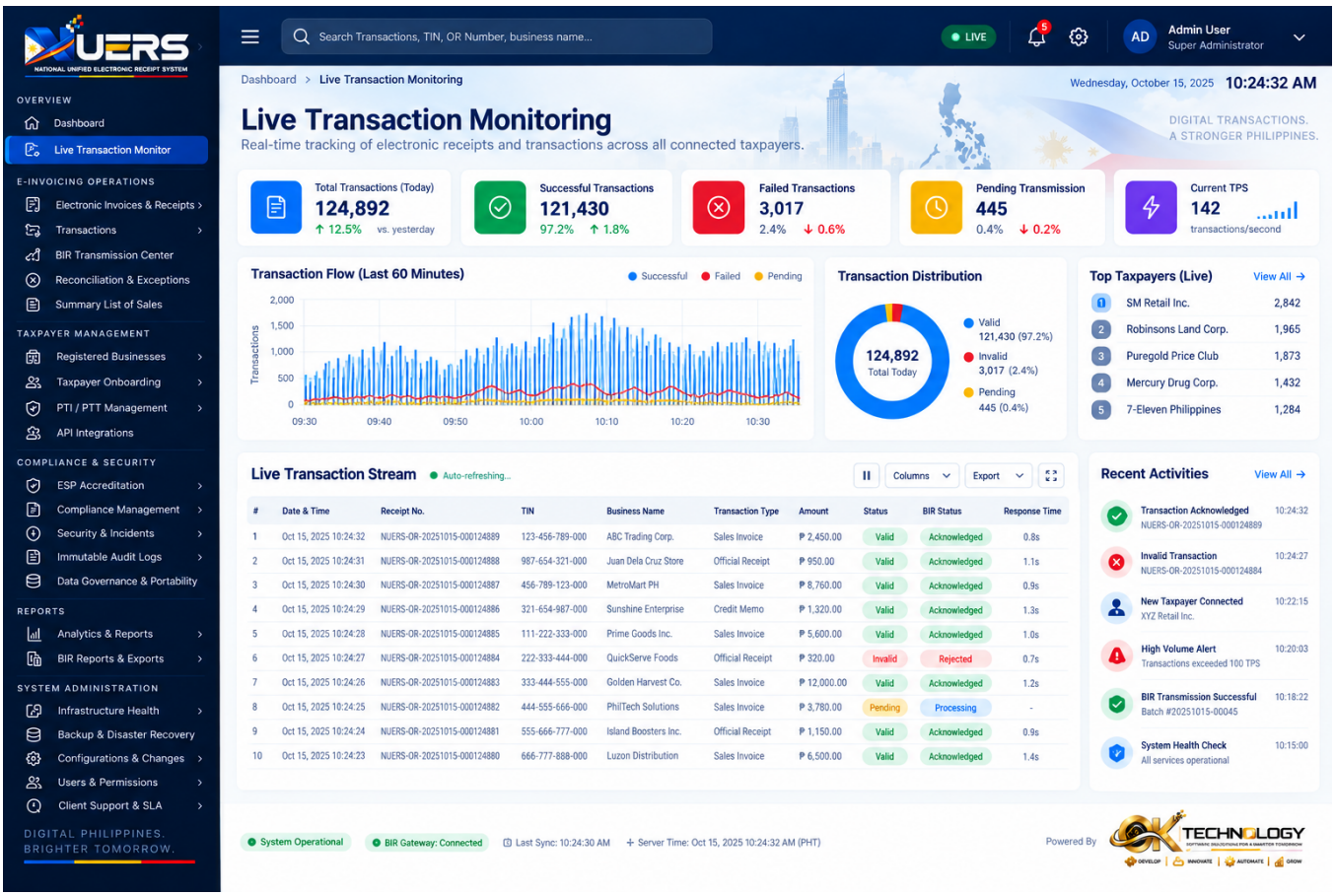
Business rules and backend logic

Counters from ledger and BIR status, not mocked client-side; clearly show last refresh.

Audit, access and QA

Enforce role and tenant/branch scope for every request. Record actor, before/after status, correlation ID, event time and evidence references. QA: empty state, invalid fields, duplicate submit, offline timeout, permissions, pagination, data consistency and safe retry.

5. Screen specification: Live Transaction Monitor



Design reference only; mock statuses and sample business data are not production evidence.

Purpose and EIS justification

Live issuance, error, queue, and throughput monitoring.

Controls, forms and modals

Pause/resume, status filters, transaction drawer, export modal.

Standard form requirements: explicit required fields, server-side validation, contextual help, disabled submit while saving, meaningful error states, accessible keyboard focus and cancellation without partial writes. Confirmation dialogs show affected tenant/document, operation, risk and reason.

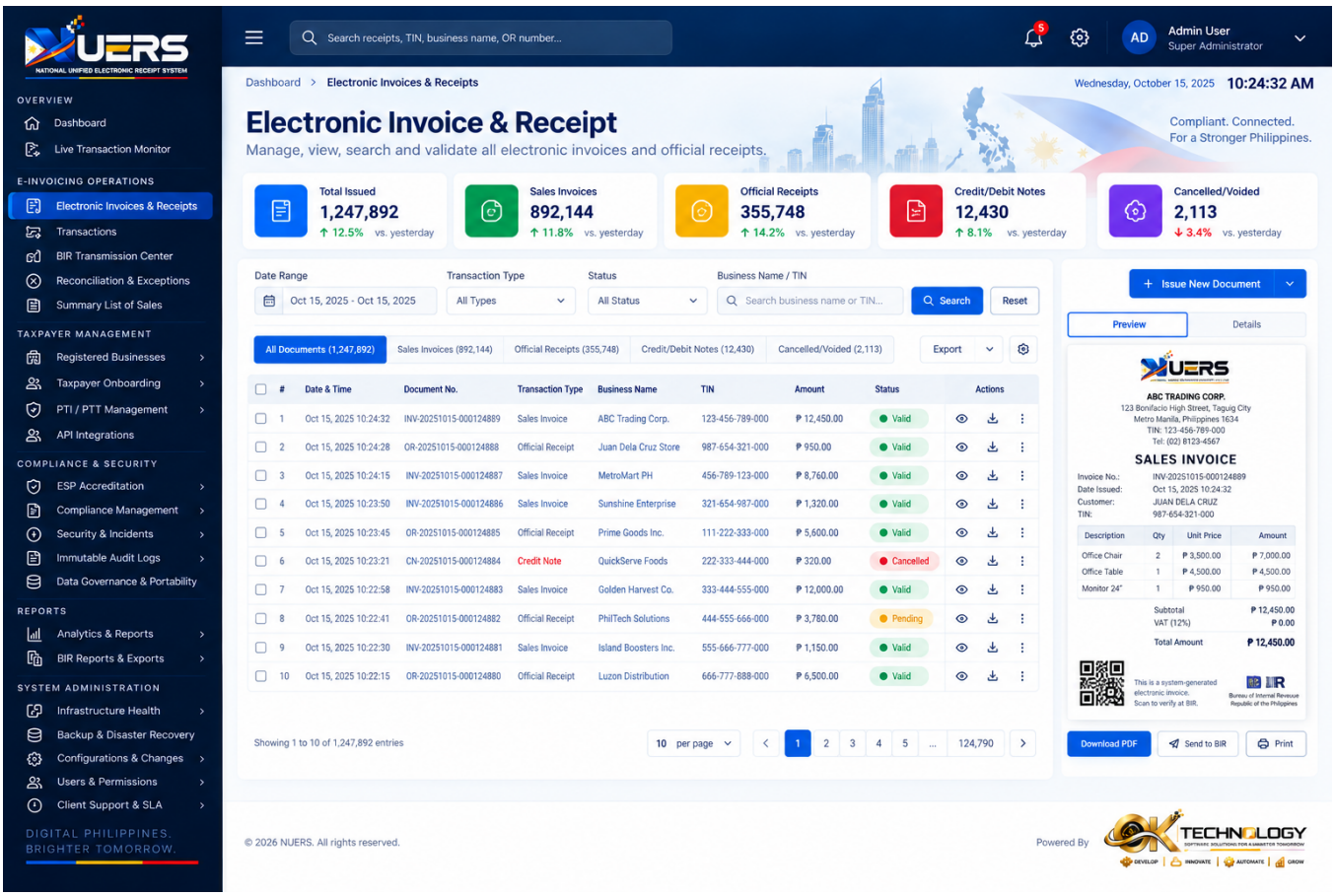
Business rules and backend logic

Cursor-based stream, reconnection, deduplication, latency and backlog telemetry.

Audit, access and QA

Enforce role and tenant/branch scope for every request. Record actor, before/after status, correlation ID, event time and evidence references. QA: empty state, invalid fields, duplicate submit, offline timeout, permissions, pagination, data consistency and safe retry.

5. Screen specification: Electronic Invoices & Receipts



Design reference only; mock statuses and sample business data are not production evidence.

Purpose and EIS justification

Manage legally classified invoices and receipt records.

Controls, forms and modals

Issue wizard, buyer/seller and line-item form, tax summary, preview, confirmation and download.

Standard form requirements: explicit required fields, server-side validation, contextual help, disabled submit while saving, meaningful error states, accessible keyboard focus and cancellation without partial writes. Confirmation dialogs show affected tenant/document, operation, risk and reason.

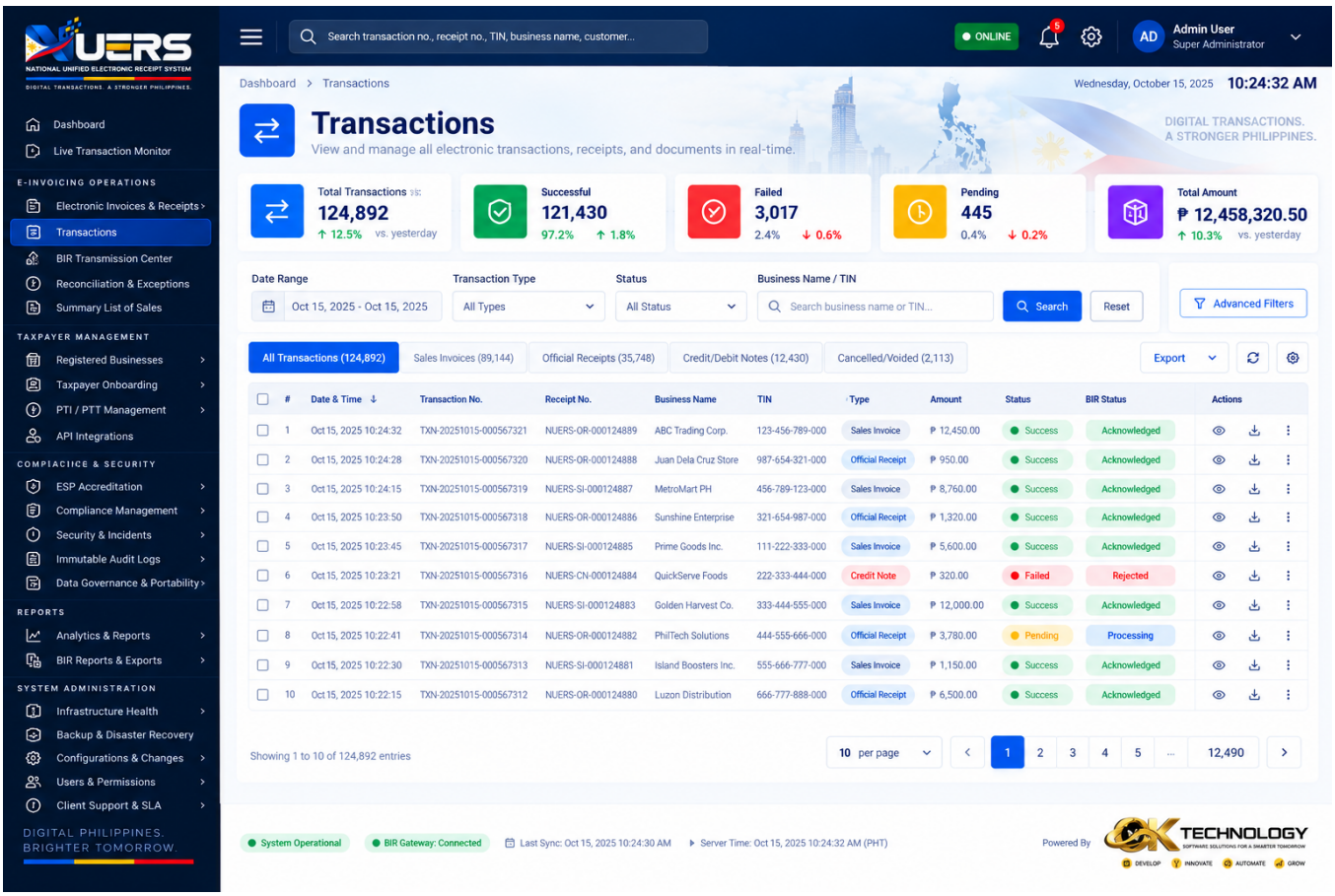
Business rules and backend logic

Validate taxpayer authority, invoice numbering, tax calculations, canonical payload and immutable issuance.

Audit, access and QA

Enforce role and tenant/branch scope for every request. Record actor, before/after status, correlation ID, event time and evidence references. QA: empty state, invalid fields, duplicate submit, offline timeout, permissions, pagination, data consistency and safe retry.

5. Screen specification: Transactions



Design reference only; mock statuses and sample business data are not production evidence.

Purpose and EIS justification

Search authoritative transaction ledger across tenants.

Controls, forms and modals

Advanced filters, detail drawer, status timeline, export request.

Standard form requirements: explicit required fields, server-side validation, contextual help, disabled submit while saving, meaningful error states, accessible keyboard focus and cancellation without partial writes. Confirmation dialogs show affected tenant/document, operation, risk and reason.

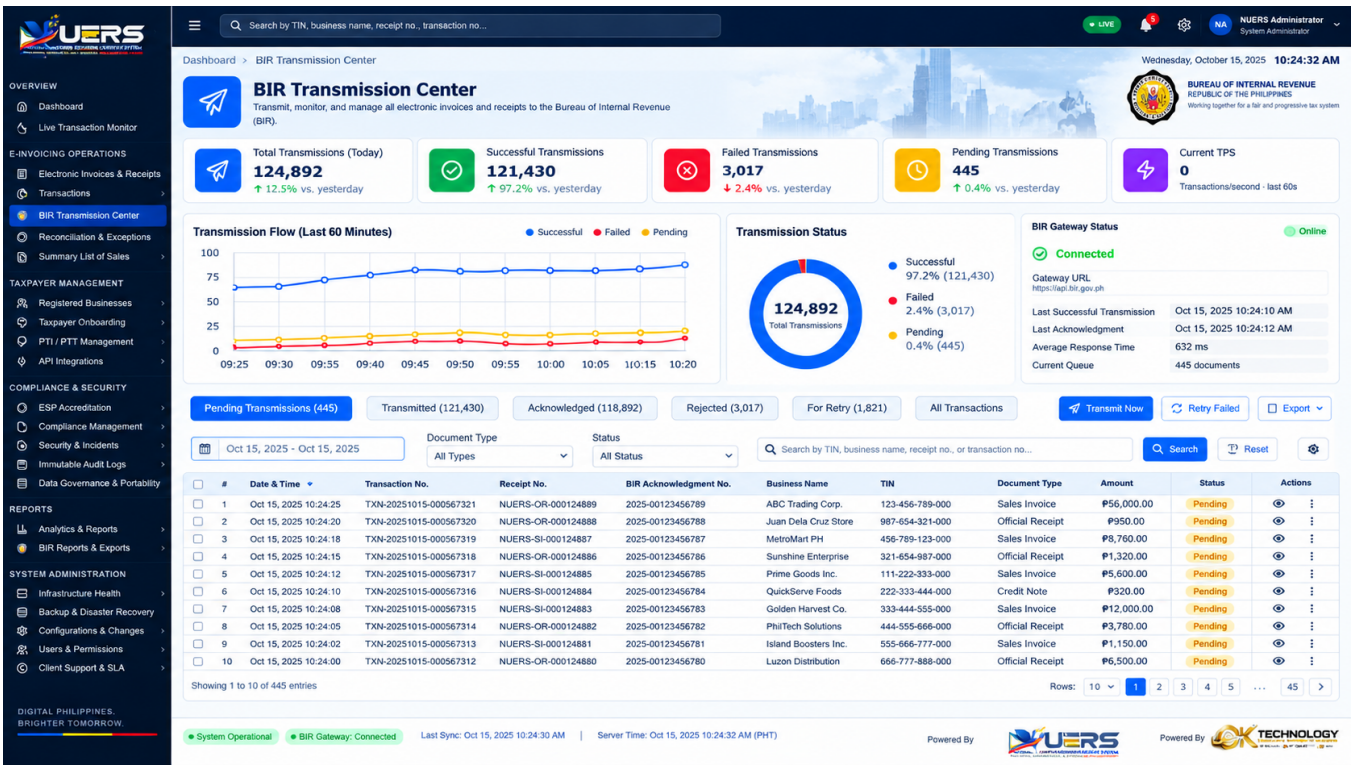
Business rules and backend logic

One transaction links to issued document, attempts, acknowledgments and audit events.

Audit, access and QA

Enforce role and tenant/branch scope for every request. Record actor, before/after status, correlation ID, event time and evidence references. QA: empty state, invalid fields, duplicate submit, offline timeout, permissions, pagination, data consistency and safe retry.

5. Screen specification: BIR Transmission Center



Design reference only; mock statuses and sample business data are not production evidence.

Purpose and EIS justification

Manage submission queue, BIR connectivity, acknowledgment and retry.

Controls, forms and modals

Queue tabs, transmission details, retry confirmation, failure details and gateway status.

Standard form requirements: explicit required fields, server-side validation, contextual help, disabled submit while saving, meaningful error states, accessible keyboard focus and cancellation without partial writes. Confirmation dialogs show affected tenant/document, operation, risk and reason.

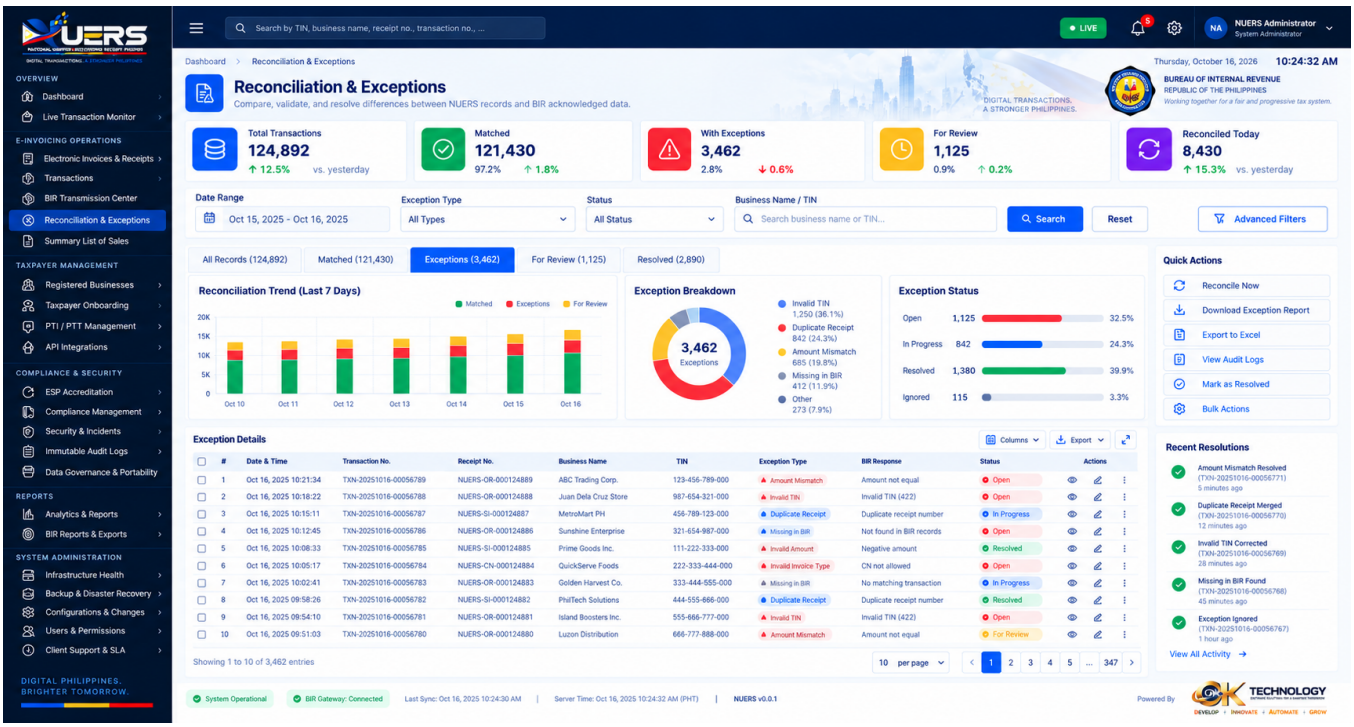
Business rules and backend logic

Durable outbox, exponential backoff, deadline alerts, unknown-result reconciliation; no duplicate submissions.

Audit, access and QA

Enforce role and tenant/branch scope for every request. Record actor, before/after status, correlation ID, event time and evidence references. QA: empty state, invalid fields, duplicate submit, offline timeout, permissions, pagination, data consistency and safe retry.

5. Screen specification: Reconciliation & Exceptions



Design reference only; mock statuses and sample business data are not production evidence.

Purpose and EIS justification

Detect missing BIR acknowledgments, mismatches, duplicate or rejected submissions.

Controls, forms and modals

Assign exception, evidence drawer, controlled resolution modal and reconciliation report.

Standard form requirements: explicit required fields, server-side validation, contextual help, disabled submit while saving, meaningful error states, accessible keyboard focus and cancellation without partial writes. Confirmation dialogs show affected tenant/document, operation, risk and reason.

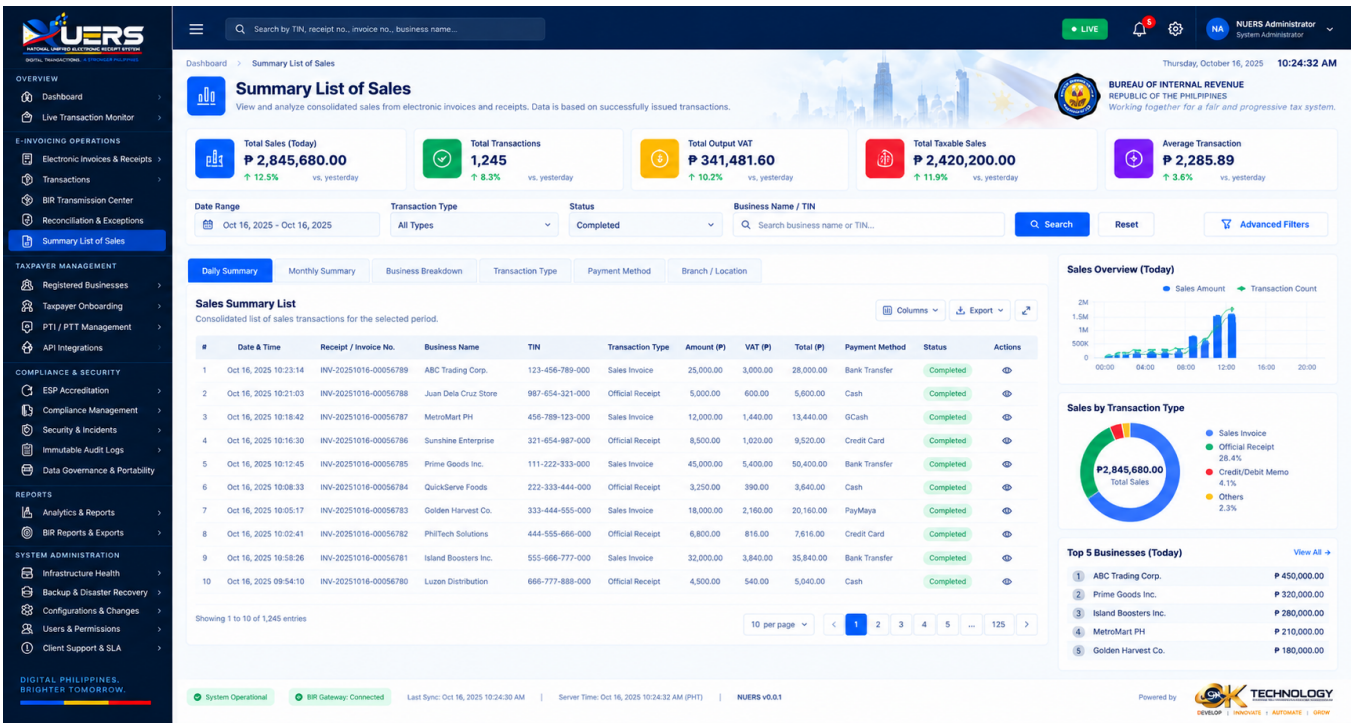
Business rules and backend logic

Compare canonical hashes and IDs; never auto-mark resolved without evidence.

Audit, access and QA

Enforce role and tenant/branch scope for every request. Record actor, before/after status, correlation ID, event time and evidence references. QA: empty state, invalid fields, duplicate submit, offline timeout, permissions, pagination, data consistency and safe retry.

5. Screen specification: Summary List of Sales



Design reference only; mock statuses and sample business data are not production evidence.

Purpose and EIS justification

Period-based sales, tax and branch summaries.

Controls, forms and modals

Date range, branch, document type, export, drilldown.

Standard form requirements: explicit required fields, server-side validation, contextual help, disabled submit while saving, meaningful error states, accessible keyboard focus and cancellation without partial writes. Confirmation dialogs show affected tenant/document, operation, risk and reason.

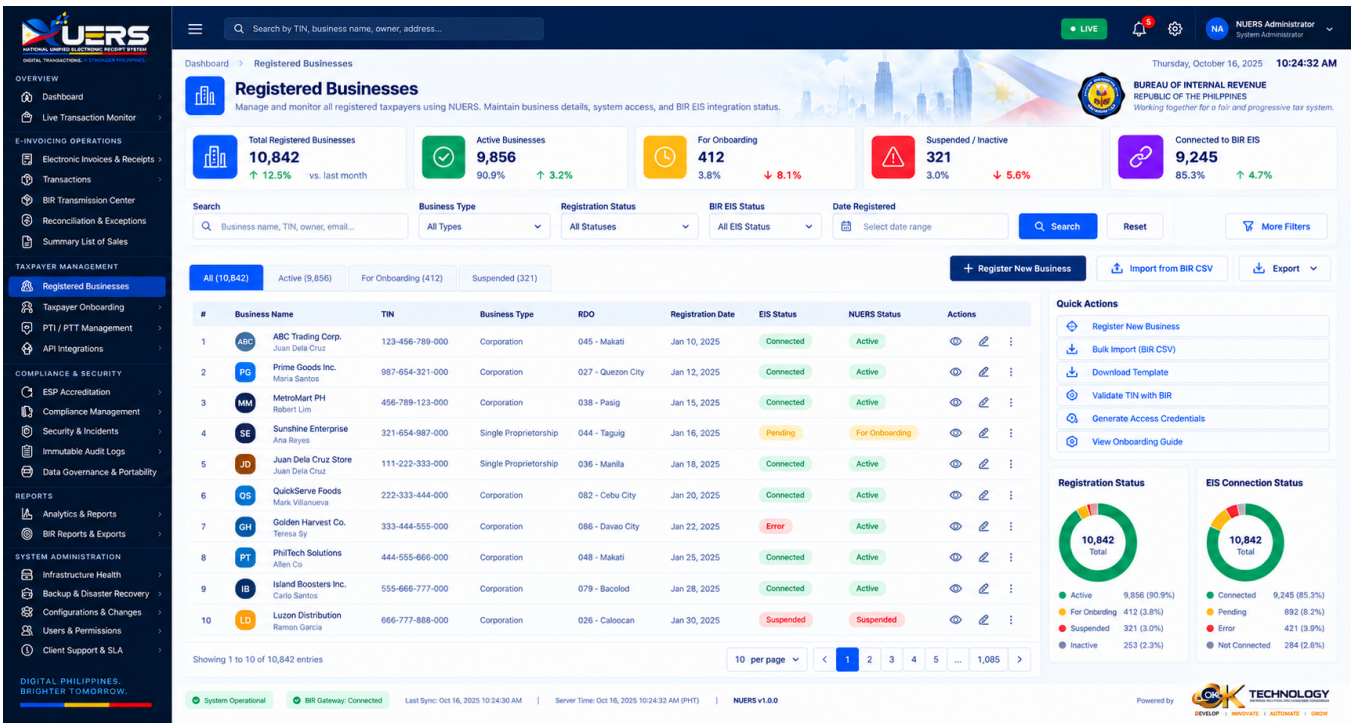
Business rules and backend logic

Exclude double counting; credit/debit and void treatment follows applicable tax rules.

Audit, access and QA

Enforce role and tenant/branch scope for every request. Record actor, before/after status, correlation ID, event time and evidence references. QA: empty state, invalid fields, duplicate submit, offline timeout, permissions, pagination, data consistency and safe retry.

5. Screen specification: Registered Businesses



Design reference only; mock statuses and sample business data are not production evidence.

Purpose and EIS justification

Manage registered taxpayer organizations and branches.

Controls, forms and modals

New business wizard, identity review, branch form, import validation and suspension confirmation.

Standard form requirements: explicit required fields, server-side validation, contextual help, disabled submit while saving, meaningful error states, accessible keyboard focus and cancellation without partial writes. Confirmation dialogs show affected tenant/document, operation, risk and reason.

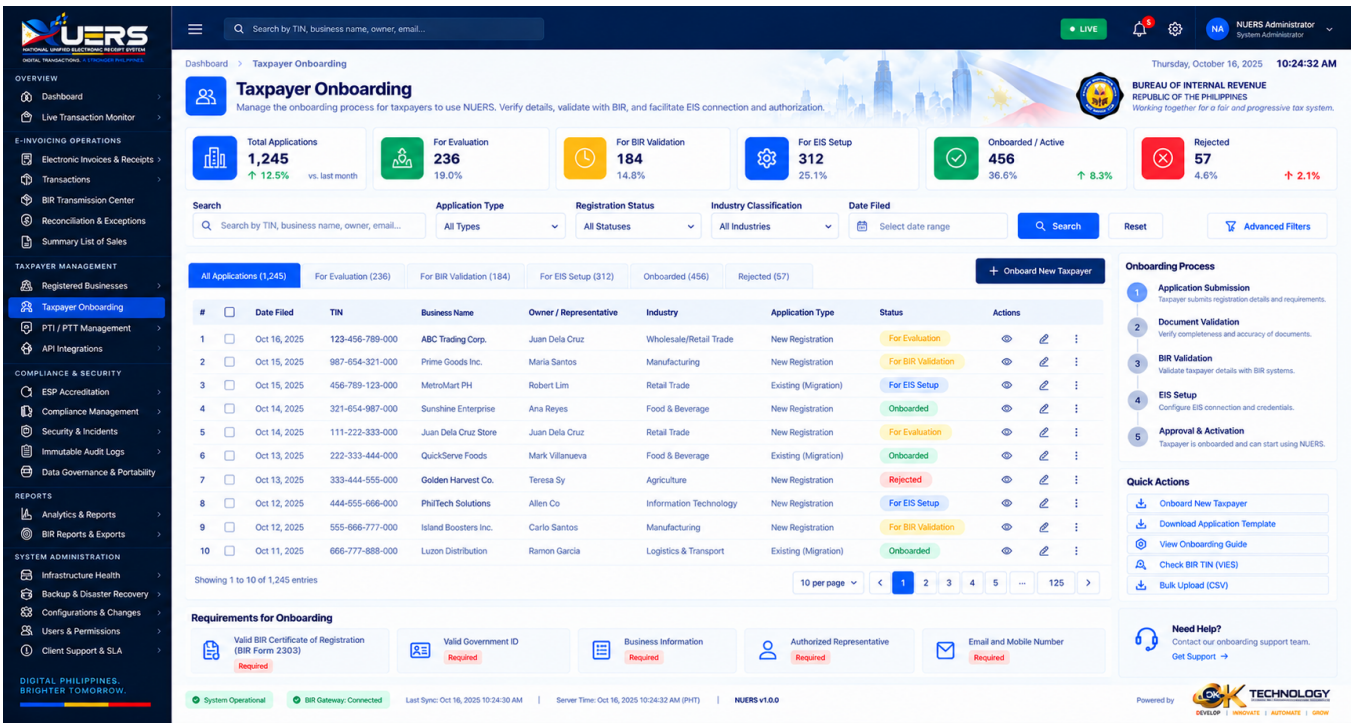
Business rules and backend logic

Distinguish NUERS account status from BIR registration and EIS eligibility.

Audit, access and QA

Enforce role and tenant/branch scope for every request. Record actor, before/after status, correlation ID, event time and evidence references. QA: empty state, invalid fields, duplicate submit, offline timeout, permissions, pagination, data consistency and safe retry.

5. Screen specification: Taxpayer Onboarding



Design reference only; mock statuses and sample business data are not production evidence.

Purpose and EIS justification

Collect evidence, review and authorize taxpayer readiness.

Controls, forms and modals

Application wizard, file upload, reviewer checklist, request changes, approval modal.

Standard form requirements: explicit required fields, server-side validation, contextual help, disabled submit while saving, meaningful error states, accessible keyboard focus and cancellation without partial writes. Confirmation dialogs show affected tenant/document, operation, risk and reason.

Business rules and backend logic

Require verified identity, approved authorization, integration tests and signoff before production.

Audit, access and QA

Enforce role and tenant/branch scope for every request. Record actor, before/after status, correlation ID, event time and evidence references. QA: empty state, invalid fields, duplicate submit, offline timeout, permissions, pagination, data consistency and safe retry.

6. PTI / PTT Management

Authorization registry, document validity, taxpayer ownership and expiration reminders. Forms: permit registration, evidence upload, review and revoke. Gate issuance and transmission separately.

Why needed: This module supports a specific EIS operational, governance, reporting or continuity obligation; exact controls are NUERS design proposals unless expressly grounded in RMC 110-2026.

Required workflow: Search/list -> create or inspect -> validate -> approval where sensitive -> persist -> notify -> audit -> export. Implement tenant scope, loading/empty/error states, status history and QA for unauthorized access and repeated submissions.

6. API Integrations

POS/ERP clients, credentials, scopes, IP policy, sandbox, webhooks, signature validation and rotation. Modals: create client, rotate/revoke key, test callback. Never expose stored secrets.

Why needed: This module supports a specific EIS operational, governance, reporting or continuity obligation; exact controls are NUERS design proposals unless expressly grounded in RMC 110-2026.

Required workflow: Search/list -> create or inspect -> validate -> approval where sensitive -> persist -> notify -> audit -> export. Implement tenant scope, loading/empty/error states, status history and QA for unauthorized access and repeated submissions.

6. ESP Accreditation

Approved solution/version/environment, certificate validity, annual review, material-change tracking. Forms: accreditation evidence, approved interfaces, renewal and controlled changes.

Why needed: This module supports a specific EIS operational, governance, reporting or continuity obligation; exact controls are NUERS design proposals unless expressly grounded in RMC 110-2026.

Required workflow: Search/list -> create or inspect -> validate -> approval where sensitive -> persist -> notify -> audit -> export. Implement tenant scope, loading/empty/error states, status history and QA for unauthorized access and repeated submissions.

6. Compliance Management

Control register, requirement-to-evidence mapping, owners, deadlines, findings and corrective action. Forms: control attestation, remediation, approval and exception request.

Why needed: This module supports a specific EIS operational, governance, reporting or continuity obligation; exact controls are NUERS design proposals unless expressly grounded in RMC 110-2026.

Required workflow: Search/list -> create or inspect -> validate -> approval where sensitive -> persist -> notify -> audit -> export. Implement tenant scope, loading/empty/error states, status history and QA for unauthorized access and repeated submissions.

6. Security & Incidents

Threat monitoring, access anomalies, vulnerabilities, severity and reporting. Forms: incident timeline, affected taxpayers, regulator notification, evidence and closure.

Why needed: This module supports a specific EIS operational, governance, reporting or continuity obligation; exact controls are NUERS design proposals unless expressly grounded in RMC 110-2026.

Required workflow: Search/list -> create or inspect -> validate -> approval where sensitive -> persist -> notify -> audit -> export. Implement tenant scope, loading/empty/error states, status history and QA for unauthorized access and repeated submissions.

6. Immutable Audit Logs

Append-only actor/system events, synchronized timestamps, tamper evidence, search, legal hold and export approvals. No delete/edit control for immutable records.

Why needed: This module supports a specific EIS operational, governance, reporting or continuity obligation; exact controls are NUERS design proposals unless expressly grounded in RMC 110-2026.

Required workflow: Search/list -> create or inspect -> validate -> approval where sensitive -> persist -> notify -> audit -> export. Implement tenant scope, loading/empty/error states, status history and QA for unauthorized access and repeated

submissions.

6. Data Governance & Portability

Data ownership, access, export, retention, migration and lawful holds. Forms: machine-readable export request, migration authorization, delivery confirmation and deletion only where lawful.

Why needed: This module supports a specific EIS operational, governance, reporting or continuity obligation; exact controls are NUERS design proposals unless expressly grounded in RMC 110-2026.

Required workflow: Search/list -> create or inspect -> validate -> approval where sensitive -> persist -> notify -> audit -> export. Implement tenant scope, loading/empty/error states, status history and QA for unauthorized access and repeated submissions.

6. Analytics & Reports

Throughput, rejection rate, queue age, branch activity, compliance performance and capacity. Filters and scheduled report forms; all figures reconcile to source ledger.

Why needed: This module supports a specific EIS operational, governance, reporting or continuity obligation; exact controls are NUERS design proposals unless expressly grounded in RMC 110-2026.

Required workflow: Search/list -> create or inspect -> validate -> approval where sensitive -> persist -> notify -> audit -> export. Implement tenant scope, loading/empty/error states, status history and QA for unauthorized access and repeated submissions.

6. BIR Reports & Exports

Prescribed reports, SLS, detailed machine-readable extracts, secure delivery and response deadlines. Forms: reporting period, format, approval and delivery audit.

Why needed: This module supports a specific EIS operational, governance, reporting or continuity obligation; exact controls are NUERS design proposals unless expressly grounded in RMC 110-2026.

Required workflow: Search/list -> create or inspect -> validate -> approval where sensitive -> persist -> notify -> audit -> export. Implement tenant scope, loading/empty/error states, status history and QA for unauthorized access and repeated submissions.

6. Infrastructure Health

Web/database/storage/queue/BIR integration health, latency, capacity and alert escalation. Forms: alert rules, incident drilldown and maintenance approval.

Why needed: This module supports a specific EIS operational, governance, reporting or continuity obligation; exact controls are NUERS design proposals unless expressly grounded in RMC 110-2026.

Required workflow: Search/list -> create or inspect -> validate -> approval where sensitive -> persist -> notify -> audit -> export. Implement tenant scope, loading/empty/error states, status history and QA for unauthorized access and repeated submissions.

6. Backup & Disaster Recovery

Encrypted backups, replication, restore tests, failover drills, manual invoicing reconciliation and RTO/RPO evidence. Forms: restore request, DR exercise and signoff.

Why needed: This module supports a specific EIS operational, governance, reporting or continuity obligation; exact controls are NUERS design proposals unless expressly grounded in RMC 110-2026.

Required workflow: Search/list -> create or inspect -> validate -> approval where sensitive -> persist -> notify -> audit -> export. Implement tenant scope, loading/empty/error states, status history and QA for unauthorized access and repeated submissions.

6. Configurations & Changes

Versioned schema, templates, feature flags, integration endpoints and material change approval. Forms: change request, risk impact, rollout and rollback.

Why needed: This module supports a specific EIS operational, governance, reporting or continuity obligation; exact controls are NUERS design proposals unless expressly grounded in RMC 110-2026.

Required workflow: Search/list -> create or inspect -> validate -> approval where sensitive -> persist -> notify -> audit -> export. Implement tenant scope, loading/empty/error states, status history and QA for unauthorized access and repeated submissions.

6. Users & Permissions

Role and branch access, MFA, invitation, deactivation and access review. Forms: invite, role assignment, credential reset, privilege elevation approval.

Why needed: This module supports a specific EIS operational, governance, reporting or continuity obligation; exact controls are NUERS design proposals unless expressly grounded in RMC 110-2026.

Required workflow: Search/list -> create or inspect -> validate -> approval where sensitive -> persist -> notify -> audit -> export. Implement tenant scope, loading/empty/error states, status history and QA for unauthorized access and repeated submissions.

6. Client Support & SLA

Ticket intake, incident escalation, taxpayer communications, service levels and evidence. Forms: ticket, escalation, outage announcement and resolution.

Why needed: This module supports a specific EIS operational, governance, reporting or continuity obligation; exact controls are NUERS design proposals unless expressly grounded in RMC 110-2026.

Required workflow: Search/list -> create or inspect -> validate -> approval where sensitive -> persist -> notify -> audit -> export. Implement tenant scope, loading/empty/error states, status history and QA for unauthorized access and repeated submissions.

7. Detailed invoice issuance form

Field group	Required fields / validation
Seller	TIN, legal name, branch, registered address, applicable tax status, verified PTI; derive from approved taxpayer record
Buyer	Name, address, TIN and other legally required fields according to invoice type; do not assume every buyer has TIN
Document	Document type, unique sequence, source reference, issuance timestamp, currency, transaction classification
Line items	Description, quantity, unit price, discount, tax treatment, computed totals
Payment	Payment references and settlement metadata where relevant; do not confuse payment status with invoice issuance
Compliance	Canonical schema version, integrity hash, permitted invoice authority, transmission eligibility

Modals: Preview before issue; Confirm Issue (irreversible issued record); Validation Errors (field-by-field); Issue Result (document ID, state and next steps); Correct/Adjust (authorized legally permitted corrective document). Never provide a silent edit to an issued invoice.

8. Detailed BIR transmission workflow

1. Select issued eligible documents. 2. Validate taxpayer and interface authorization. 3. Serialize to BIR-approved schema. 4. Commit delivery intent and hash. 5. Submit through officially provisioned secure interface. 6. Record request/response and latency. 7. Parse authoritative acceptance/rejection. 8. Update state and reconciliation. 9. On timeout, mark uncertain and inquire before resubmission. 10. Preserve all attempts and notify owners on deadline risk.

Transmission modals: Batch details with count and scope; retry confirmation with last response; response viewer with redacted secrets; incident/outage declaration; manual reconciliation approval. The official endpoint, response codes and payload schema remain pending authoritative BIR documentation.

9. Security, performance and disaster recovery

Proposed infrastructure: two 64-physical-core/128-GB Web nodes; two similarly sized database nodes plus independent quorum/fencing mechanism; three 64-core/128-GB object-storage nodes. Separate durable queue, monitoring, immutable backups and offsite DR. Two DB nodes alone are insufficient for quorum-safe automatic promotion.

Target load: 1 million transactions/hour (~278 TPS average), with burst testing to 2,000 TPS; these are design targets, not proven capacity. Test complete invoice issuance, persistence, acknowledgments, audit and document delivery. Provide WAF, TLS, encryption at rest, MFA, access controls, security review, penetration testing and monitored incident response.

10. Production acceptance gates

Gate A: Confirm ESP eligibility and solution-specific accreditation path. Gate B: Confirm taxpayer PTI/PTT authority and applicable EIS requirements. Gate C: Obtain official interface specifications and complete conformance tests. Gate D: Validate accounting/tax rules and invoice numbering. Gate E: Complete load, security, failover, backup/restore, privacy and interoperability tests. Gate F: Document BIR-supervised tests and approvals before live taxpayer production.

Acceptance cases: 1M/hour sustained, one web node loss, database promotion without lost acknowledged writes, one storage node failure, duplicate source replay, BIR 5xx/timeouts, acknowledgment mismatch, near-deadline queue, taxpayer permit expiry, tenant cross-access attempt, manual outage invoice linkage, full restore, and structured export on termination.

11. Regulatory traceability and outstanding decisions

Source-derived: RMC No. 110-2026 pages 2-12, covering ESP scope, accreditation, issuance/transmission distinctions, security, performance, obligations, taxpayer data rights and liability. **Design-derived:** NUERS screens, workflow names, server topology, tables, API abstraction, queue architecture, status names, field groups and UI modals.

Open items: official EIS schema and API documentation; document tax classifications; PTI/PTT issuance process; authorized BIR sandbox and credentials; accreditation evidence checklist/Annex A; retention and residency interpretation; detailed service-level metrics; approved BIR logo usage and public representation.